HERIOT-WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

DEPARTMENT OF ACTUARIAL MATHEMATICS AND STATISTICS

F78AA1

ACTUARIAL AND FINANCIAL MATHEMATICS A

18 December 2008

Time allowed 2 hours

Attempt all 7 questions

Total Marks 60

Only approved calculators may be used

1. (a) Prove from first principles that:

$$s^{i}_{\overline{n}|} = \frac{(1+i)^n - 1}{i}.$$

[3 marks]

(b) Evaluate the following at interest of 10% *per annum* effective:

(i) $\ddot{s}_{\overline{20}|}$

(ii) $a_{\overline{\infty}|}$

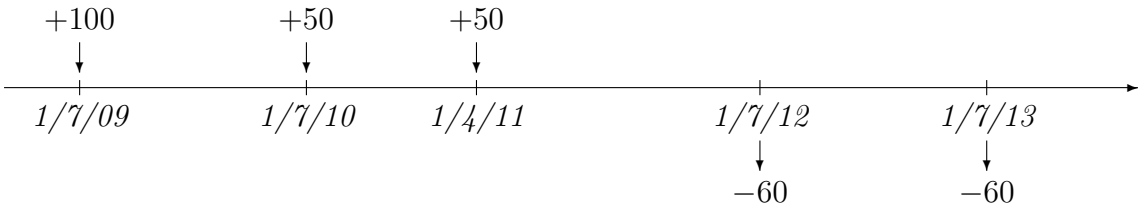
(iii) $\ddot{a}^{(4)}_{\overline{30}|}$

(iv) ${}_{10}|a_{\overline{10}|}$.

[8 marks]

[Total 11 marks]

2. The effective rate of interest will be 4% per half-year during 2009 and 2010, and 9% *per annum* thereafter. A series of cashflows is given as follows:



(a) What is the present value on 1 January 2009 of these cashflows?

(b) What is the accumulation on 1 January 2014 of these cashflows?

[6 marks]

continued ...

3. You borrow £25,000 to build an extension to your house. The loan is to be repaid over 10 years by level monthly repayments, payable in arrear. The interest rate is initially 6.16778% *per annum* effective, but can be changed by the lender at any time.

   (a) (i) What is the level monthly repayment?
   (ii) What is the capital outstanding just after the 72nd repayment has been made?

   [4 marks]

   (b) Just after the 72nd repayment has been made, the lender increases the interest rate to 8.73107% *per annum* effective. However, you cannot afford to pay more each month than you are currently paying. You ask the lender to extend the outstanding term of the loan, so that it will be repaid as soon as possible, with the level monthly repayment not exceeding its current amount. By how many months will the term of the loan be extended?

   [6 marks]

   [Total 10 marks]

4. A bond has remaining term 7.5 years, with coupons of 8% *per annum* payable half-yearly in arrear. The coupon due now has just been paid. The bond is redeemable at a discount of 2%. Its market price is such that if an investor subject to income tax at 25% and capital gains tax at 40% were to purchase it, he would obtain a net redemption yield of 12% *per annum* effective. What is the price per £100 nominal of the bond?

   [6 marks]

5. An investor is considering investing in a proposal to set up and run a new organic market garden. Buying the land and preparing the site will cost £1 million immediately. Certification of organic status takes two calendar years, during which there will be no income. At the end of each of the third to sixth calendar years, inclusive, net income of £500,000 will be received. At the end of the sixth calendar year, the land will be sold for £200,000.

   (a) Show that the Net Present Value of the project, at at 12% *per annum* effective, is £312,004.

   [2 mark]

   (b) What is the Internal Rate of Return on the project? [6 marks]

   (c) If the investor can borrow money at 6% *per annum* effective, and deposit money to earn 4% *per annum* effective, what profit will he make on the project at the end of the six years?

   [4 marks]

   [Total 12 marks]

continued ...

6. On 1 June 2008 an investor bought a U.K. Government 3% Index-Linked Stock at a price of £110.00 per £100 nominal.

   - The base RPI figure for indexing is 182.3.
   - The most recent RPI is for April 2008 and was 214.0.
   - Coupons are payable twice yearly on 1 June and 1 December.
   - The bond will be redeemed on 1 June 2018.
   - The coupon for June 2008 has already been paid.
   - Coupons and the final redemption payment are indexed in line with the RPI with a time lag of 8 months.

   (a) Calculate the amount of the next coupon, due on 1 December 2008.

   [2 marks]

   (b) Assuming that RPI will grow continuously at the rate of 2.5% *per annum*, calculate the projected amount of the final payment (coupon plus redemption proceeds) on 1 June 2018.

   [3 marks]

   (c) Show that, with RPI as in (b) above, the monetary yield on this index-linked bond lies between 6% and 7% per annum effective.

   [5 marks]

   (d) Current economic conditions suggest that RPI will be higher than 2.5% *per annum* for a short while. Assume that RPI will grow continuously at the rate of 4% per annum from May 2008 to April 2010 inclusive, and thereafter at 2.5% *per annum*. Without doing any further calculations, state with reasons whether the monetary yield on the bond will decrease, stay the same or increase.

   [2 marks]

   [Total 12 Marks]

7. On 31 December 2005, an investment fund has assets worth £100,000. On 1 January 2006 it pays out £20,000, on 1 January 2007 it receives an inflow of £20,000, and on 31 December 2007 it has assets worth £110,000. On 1 January 2008 it pays out £20,000, and on 31 December 2008 it has assets worth £100,000. Find the linked internal rate of return during the years 2006–2008, based on the subdivision of the period into two parts, one being the calendar years 2006–2007 and the other being the calendar year 2008.

   [6 marks]

Total 60 marks